Christian Covington

Department of Biostatistics, Harvard T.H. Chan School of Public Health
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EDUCATION

Ph.D. in Biostatistics 2022 – 2027 (Expected)
Harvard T.H. Chan School of Public Health, Boston, MA
Dissertation: *A formal framework for Bayesian model criticism with frequentist guarantees*.

MMath in Computer Science 2020 – 2022
University of Waterloo, Waterloo, ON, Canada

B.A. in Statistical Science 2014 – 2016
Cornell University, Ithaca, NY

RESEARCH INTERESTS

Model misspecification, Model uncertainty, Bayesian statistics, Foundations of statistical inference

PUBLICATIONS

Works in progress

Chattopadhyay, S., C. T. Covington, and J. W. Miller (n.d.). *Uniform parametrization checks for Bayesian nonparametric models*.

Covington, C. T., M. B. Mathur, and T. J. VanderWeele (n.d.). *Multiverse analysis for causal inference (and vice-versa)*.

Covington, C. T. and J. W. Miller (n.d.). *Canonical representations of u-values for Bayesian model criticism via the generalized Rosenblatt transform*.

Statistical Theory & Methodology

Covington, C. T. and J. W. Miller (2025a). *A powerful goodness-of-fit test using the probability integral transform of order statistics*. arXiv: `2510.22854 [stat.ME]`. Under review.

Covington, C. T. and J. W. Miller (2025b). *Bayesian model criticism using uniform parametrization checks*. arXiv: `2503.18261 [stat.ME]`. Under review.

Mathur, M. B., C. T. Covington, and T. J. VanderWeele (2023). "Variation across analysts in statistical significance, yet consistently small effect sizes". In: *Proceedings of the National Academy of Sciences* 120.3, e2218957120.

Covington, C., X. He, J. Honaker, and G. Kamath (2021). "Unbiased statistical estimation and valid confidence intervals under differential privacy". In: *Statistica Sinica*.

Collaborative & Applied Research

Song, N., M. Frean, C. T. Covington, M. Tietschert, E. Ling, and H. Bahadurzada (2022). "Patients' perceptions of integrated care among Medicare beneficiaries by level of need for health services". In: *Medical Care Research and Review* 79.5, pp. 640–649.

Frean, M., C. Covington, M. Tietschert, H. Bahadurzada, J. So, and S. J. Singer (2021). "Patient experiences of integrated care in Medicare Accountable Care Organizations and Medicare Advantage versus traditional Fee-for-Service". In: *Medical Care* 59.3, pp. 195–201.

Ling, E. J., M. Frean, J. So, M. Tietschert, N. Song, and C. Covington (2021). "Differences in patient perceptions of integrated care among Black, Hispanic, and White Medicare beneficiaries". In: *Health Services Research* 56.3, pp. 507–516.

Jiang, X., A. B. Hall, T. D. Arthur, D. R. Plichta, C. T. Covington, and M. Poyet (2019). "Invertible promoters mediate bacterial phase variation, antibiotic resistance, and host adaptation in the gut". In: *Science* 363.6423, pp. 181–187.

Teaching/Mentoring Experience

Teaching Assistant

BST 240: Probability II 2024, 2025
Harvard T.H. Chan School of Public Health

BST 210: Applied Regression Analysis 2023
Harvard T.H. Chan School of Public Health

SOC 2580: Six Pretty Good Books 2016
Cornell University

Mentor

Betania Adane (High School Student) 2024
Harvard Undergraduate OpenBio Laboratory Summer Research Institute
Next position: Undergraduate at Stanford University

Tutor

Statistics Tutor 2015 – 2016
Cornell University

Reviewing Experience

- **Journals:** Philosophical Transactions of the Royal Society A
- **Conferences:** ICML, NeurIPS

Professional Experience

BridgeBio 2025 – 2026
Research Extern
Palo Alto, CA

Apple 2021
Summer Intern
Cupertino, CA

Harvard University Privacy Tools Project 2019 – 2020
Research Fellow
Cambridge, MA

Broad Institute 2018 – 2019
Associate Computational Biologist
Cambridge, MA

Harvard Laboratory for Systems Medicine 2017 – 2018
Data Scientist
Cambridge, MA

Harvard Business School 2016 – 2017
Research Associate
Boston, MA

AWARDS

Distinction in Teaching 2023, 2024
Harvard University

Graduate Excellence Award 2020 – 2022
University of Waterloo

David R. Cheriton Graduate Scholarship 2020 – 2022
University of Waterloo

Dr. Derick Wood Graduate Scholarship 2020 – 2021
University of Waterloo

Hunter R. Rawlings III Presidential Research Scholar 2014 – 2016
Cornell University